

1. Two interpretations of uncertainty

Both approaches use probability calculus and a **sampling model** $\mathbf{X} \sim p(\mathbf{x} | \pi)$. Here π is a model parameter, $\mathbf{X}$ denotes random data, and $\mathbf{x}$ is the observed dataset.

Frequentist inference treats the unknown parameter as fixed. Probability describes variation in data, estimators and interval procedures across hypothetical repetitions of the same experiment. One can evaluate a procedure at different proposed parameter values without assigning a probability distribution to the parameter itself.

Bayesian inference assigns a probability distribution to an unknown parameter to represent uncertainty, conditional on a model and prior information. The posterior updates that uncertainty after observing data. This is compatible with the parameter having one fixed physical value; assigning a distribution need not imply that the physical value changes between experiments.

An estimate with error bars is incomplete without specifying **how the interval was constructed and what its probability means**. The same numerical endpoints can support different interpretations.

2. Sampling distribution, likelihood and posterior

The following objects answer different questions:

$$p(\hat{\pi} | \pi), \quad \mathcal{L}(\pi; \mathbf{x}) = p(\mathbf{x} | \pi), \quad p(\pi | \mathbf{x}).$$

The **sampling distribution** describes estimates from repeated data at fixed π . The **likelihood** is a function of π for fixed observed data; it is not generally a probability density over π . The **posterior** is a density over π conditional on the observed data and the assumed model and prior.

Bayes' theorem gives

$$p(\pi | \mathbf{x}) = \frac{p(\mathbf{x} | \pi)p(\pi)}{p(\mathbf{x})}, \quad p(\mathbf{x}) = \int p(\mathbf{x} | \pi)p(\pi) d\pi.$$

Here $p(\pi)$ is the **prior**. For a proper prior, $p(\mathbf{x})$ is the **evidence**, also called the **prior predictive density**. The posterior is defined when this normalising integral is finite and positive.

Prior influence depends on how informative the data are about the parameter, the prior's concentration and the model's identifiability. There is no universal sample size beyond which priors cease to matter.

The maximum-likelihood and maximum-a-posteriori estimates are

$$\hat{\pi}_{\text{ML}} = \arg \max_{\pi} \mathcal{L}(\pi; \mathbf{x}), \quad \hat{\pi}_{\text{MAP}} = \arg \max_{\pi} p(\pi | \mathbf{x}).$$

They coincide for a flat prior on the relevant parameter range, provided the likelihood maximum lies within that range. Neither a point estimate nor the location of a posterior peak determines an interval's probability or coverage.

3. Frequentist confidence intervals

Let $q = 1 - \alpha$ be the desired confidence level. A confidence procedure maps a dataset to a set $C_q(\mathbf{X})$ and satisfies

$$\mathbb{P}_{\pi}[\pi \in C_q(\mathbf{X})] \geq q \quad \text{for every allowed } \pi.$$

Equality gives **exact nominal coverage**; larger coverage is **conservative**. Approximate procedures may only approach the target coverage under stated conditions.

The parameter is fixed, while the interval varies with the data. For a 95% procedure with exact coverage, about 95% of intervals from many independent repetitions contain that fixed parameter.

After observing $\mathbf{x}$, the interval $C_q(\mathbf{x})$ either contains the fixed truth or does not. The confidence level describes the *procedure*; it does not by itself assign posterior probability q to that particular interval.

4. Bayesian credible regions

A q -credible region $B_q(\mathbf{x})$ satisfies

$$\mathbb{P}[\pi \in B_q(\mathbf{x}) | \mathbf{x}] = \int_{B_q(\mathbf{x})} p(\pi | \mathbf{x}) d\pi = q.$$

This is a probability statement about the parameter conditional on the observed data, model and prior. It does not guarantee coverage q over repeated data at every fixed true parameter.

The lecture uses **highest-posterior-density (HPD) regions**. For the posteriors considered here, these have the form

$$B_q(\mathbf{x}) = \{\pi : p(\pi | \mathbf{x}) \geq k_q(\mathbf{x})\},$$

where the threshold is chosen to enclose posterior mass q . HPD regions select the largest density values and can be disconnected for a multimodal posterior. They need not coincide with equal-tailed intervals, which leave probability $(1 - q)/2$ in each tail.

Two distinct questions: how much posterior mass does this observed region contain, and how often would this region-generating procedure include a fixed truth in repeated experiments?

5. Assessing repeated-sampling coverage

For a Bayesian region rule, define its coverage at a fixed truth by

$$c_q(\pi_0) = \mathbb{P}_{\mathbf{X}|\pi_0}[\pi_0 \in B_q(\mathbf{X})].$$

Simulate M independent datasets at π_0 , repeat the complete analysis, and count the regions containing π_0 :

$$\hat{c}_q = \frac{1}{M} \sum_{r=1}^M \mathbf{1}\{\pi_0 \in B_q(\mathbf{x}^{(r)})\}, \quad \widehat{\text{SE}}(\hat{c}_q) = \sqrt{\frac{\hat{c}_q(1 - \hat{c}_q)}{M}}.$$

At $M = 10,000$ and coverage near 68%, the Monte Carlo standard error is about 0.5 percentage points. This is uncertainty in the *estimated coverage*, not the uncertainty of an individual parameter estimate. The same check applies to confidence procedures.

Lecture 3: Bayesian & frequentist probability

Linear and nonlinear examples; when the approaches agree

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6. Linear Gaussian model: an exact agreement

Observe two independent measurements with known noise variance:

$$\mathbf{X} \mid \pi \sim \mathcal{N}_2\left(\begin{bmatrix} \pi \\ \pi \end{bmatrix}, \sigma^2 I_2\right).$$

The scalar parameter has the MLE and sampling distribution

$$\hat{\pi} = \frac{X_1 + X_2}{2}, \quad \hat{\pi} \mid \pi \sim \mathcal{N}(\pi, \sigma^2/2).$$

Write $s = \sigma/\sqrt{2}$ and $z_q = \Phi^{-1}((1+q)/2)$, where Φ is the standard-normal CDF. Then an exact frequentist confidence interval is

$$C_q(\mathbf{x}) = [\hat{\pi}(\mathbf{x}) - z_q s, \hat{\pi}(\mathbf{x}) + z_q s].$$

For the formal flat prior $p(\pi) \propto 1$ on $\mathbb{R}$, completing the square gives the proper posterior

$$\pi \mid \mathbf{x} \sim \mathcal{N}(\hat{\pi}(\mathbf{x}), s^2).$$

Its HPD and central equal-tailed intervals have the same endpoints as C_q . They have both posterior mass q and fixed-parameter coverage q , for different reasons. At $q = 0.683$, $z_q \approx 1$; the parameter uncertainty is $\sigma/\sqrt{2}$, not the noise σ of one measurement.

The flat prior on $\mathbb{R}$ is **improper**: one cannot draw parameters from it. A proper uniform prior on $[-10, 10]$ instead gives a truncated Gaussian posterior. The formulas above then apply approximately when the prior boundaries have negligible influence.

7. Nonlinear Gaussian model: different coverage

Keep the same noise but change the predicted mean:

$$\mathbf{X} \mid \pi \sim \mathcal{N}_2\left(\begin{bmatrix} \pi \\ \pi^3 \end{bmatrix}, \sigma^2 I_2\right).$$

For the lecture's example, $\sigma = 0.5$ and $\pi \sim \text{Uniform}(-10, 10)$. Define

$$Q(\pi; \mathbf{x}) = \frac{(x_1 - \pi)^2 + (x_2 - \pi^3)^2}{\sigma^2}.$$

The posterior is

$$p(\pi \mid \mathbf{x}) \propto \exp[-Q(\pi; \mathbf{x})/2] \mathbf{1}_{[-10, 10]}(\pi).$$

Gaussian data noise does not imply a Gaussian posterior in π . The mean traces a curved path in data space, and the posterior's shape changes with the observation. A Gaussian width or a fixed offset from the mode need not describe its HPD region.

In the lecture's 10,000 simulated experiments at $\pi_0 = 0$, the 68.3% HPD regions cover the truth in $(61.1 \pm 0.5)\%$ of experiments. This is a result for the specified model, prior, truth and HPD rule; it is not a universal Bayesian coverage rate. Bayes' theorem still gives 68.3% posterior mass to every correctly computed region.

8. Calibration averaged over the prior

Now change the simulation experiment: draw a new truth from the *same proper prior used in the analysis*, then draw its data:

$$\pi \sim p(\pi), \quad \mathbf{X} \mid \pi \sim p(\mathbf{x} \mid \pi).$$

If every reported region has posterior mass q , averaging over this joint distribution gives

$$\begin{aligned} \mathbb{P}[\pi \in B_q(\mathbf{X})] &= \int p(\mathbf{x}) \underbrace{\int_{B_q(\mathbf{x})} p(\pi \mid \mathbf{x}) d\pi}_{q} d\mathbf{x} \\ &= q. \end{aligned}$$

Equivalently, the fixed-parameter coverage function obeys

$$\int c_q(\pi) p(\pi) d\pi = q.$$

This is **calibration averaged over the prior**. It does not require $c_q(\pi) = q$ separately at every parameter value. Undercoverage at some values can coexist with overcoverage at others. The identity requires the simulation prior and sampling model to match those used to calculate the posterior.

For the nonlinear example, drawing truths from the uniform prior gives $(68.0 \pm 0.5)\%$ coverage in the lecture simulation, consistent with 68.3%. This is compatible with the 61.1% result at the particular fixed truth $\pi_0 = 0$.

9. Bernstein–von Mises: large-sample agreement

Let $\mathbf{x}_{1:n}$ denote n i.i.d. observations from a correctly specified model with a fixed, finite-dimensional parameter. Under suitable regularity conditions, including identifiability, an interior true parameter π_0 , nonsingular Fisher information and a prior density positive and continuous near π_0 , the posterior is asymptotically Gaussian around the MLE:

$$p(\pi \mid \mathbf{x}_{1:n}) \approx \mathcal{N}(\pi; \hat{\pi}_{\text{ML}}, F_n(\pi_0)^{-1}), \quad F_n = nF_1.$$

Here F_n is information in the whole dataset. The theorem concerns the *conditional posterior* for one observed dataset as its size increases. It is distinct from a CLT or asymptotic normality of an estimator over repeated datasets.

The posterior and the sampling distribution of a regular efficient MLE then have matching asymptotic uncertainty scales. Consequently, usual central credible intervals and efficient frequentist intervals agree to first order and attain nominal coverage asymptotically at regular interior truths.

Limits. This is an asymptotic result, not an identity for arbitrary intervals or finite samples. Boundaries, weak identifiability, an unsuitable prior or a misspecified likelihood can prevent the usual agreement. Increasing the dataset does not correct an incorrect noise model. Check both posterior calculations and repeated-sampling performance for the actual analysis.